Mean-field solution of structural balance dynamics in nonzero temperature

This presentation explores the dynamics of structural balance in social networks, focusing on the impact of temperature on the stability and evolution of these networks. We will delve into the theoretical framework, model, and simulation results to understand how temperature influences the balance of relationships within a network.

by Ali Khabazian - 402416066

Supervisor: Dr. Reza Jafari



Titles:

- introduction
- model and material
- Hamiltonian and mean field solution
- simulation and algorithm
- conclusion



introduction

Fully Connected Network

We begin by considering a fully connected network, where every node is directly connected to every other node. This simplifies the analysis and allows us to focus on the fundamental principles of structural balance.

Balance Theory

Balance theory states that a network is balanced if all triads (groups of three nodes) are balanced. A triad is balanced if all three relationships are positive (+++), or if two are positive and one is negative (+--). An imbalanced triad has two negative relationships and one positive (---, ++-).

Balanced and Imbalanced Triads

The table below summarizes the different types of triads and their balance status:

Triad	Balance
+++	Balanced
+--	Balanced
---	Imbalanced
++-	Imbalanced

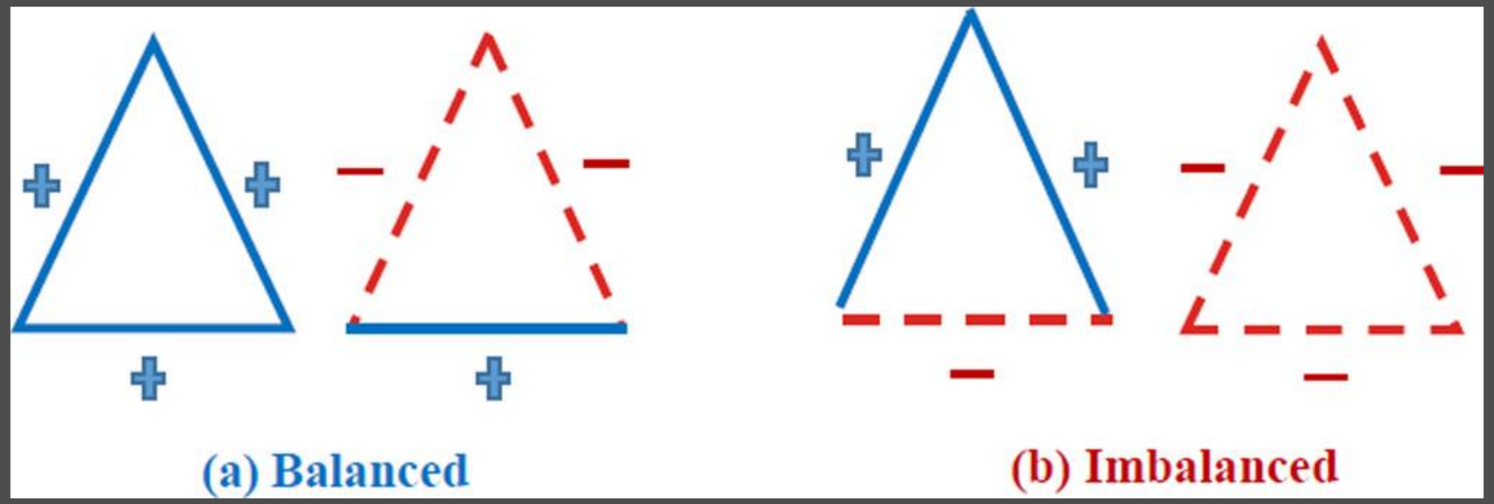


FIG. 1. Different types of triads according to structural balance.

model and material

1 Effects of Temperature

Temperature plays a crucial role in the dynamics of social networks. It represents the level of tolerance or flexibility in relationships. At low temperatures, individuals are more rigid in their opinions and less likely to change their relationships.

2 Mechanism of Structural Balance Dynamics

Structural balance dynamics are governed by the principle that individuals strive to maintain a balanced state in their relationships. This involves minimizing the number of imbalanced triads in the network.

3 Boltzmann Distribution and Imbalanced Triads

The Boltzmann distribution describes the probability of a system being in a particular state, given its energy. In the context of structural balance, the energy of a network is related to the number of imbalanced triads.

Eq1

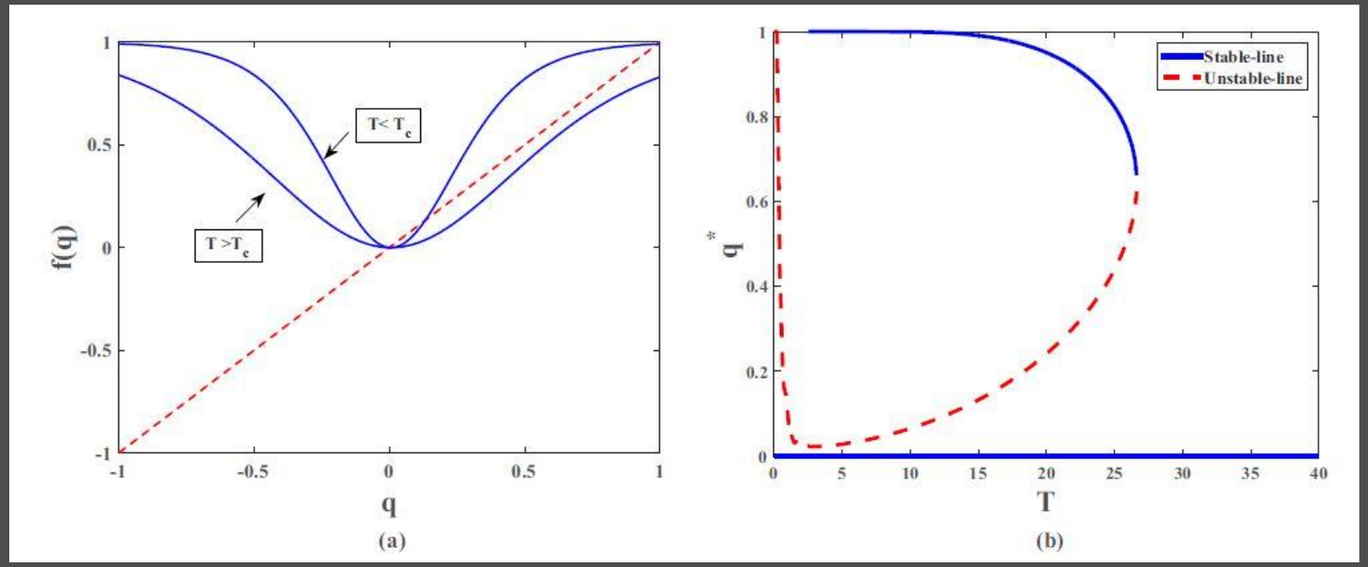


FIG. 2.
(a) Graphical analysis of Eq. (1) for $N = 50$ and in different temperatures ($T = 15, 28$).
When $T > Tc$ ($Tc = 26.2$) there is a single stable fixed point. When $T < Tc$ there are three fixed points, and the inner one is unstable. The value of T determines the number of fixed points.

(b) Bifurcation diagram showing the fixed points ($q*$) as a function of temperature (T).

$$q = \frac{e^{-\beta(N-3)(-2q)} - 2e^{-\beta(2p)} + e^{-\beta(N-3)(2q)}}{e^{-\beta(N-3)(-2q)} + 2e^{-\beta(2p)} + e^{-\beta(N-3)(2q)}}$$

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Eq. (1)

model and material

4 Phase Transition in Structural Balance Model

As temperature increases, the probability of finding the network in a balanced state decreases. At a critical temperature, the network undergoes a phase transition, where it becomes dominated by imbalanced triads.

$$\langle S_{ik}S_{kj}S_{ji}\rangle = \frac{e^{\beta(N-3)(3q)} + \beta - 3e^{\beta(N-3)q} - \beta + 3e^{\beta(N-3)(-q)} + \beta - e^{\beta(N-3)(-3q)} - \beta}{e^{\beta(N-3)(3q)} + \beta + 3e^{\beta(N-3)q} - \beta + 3e^{\beta(N-3)(-q)} + \beta + e^{\beta(N-3)(-3q)} - \beta}$$

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Eq. (2): The mean triad energy equation

Hamiltonian and mean field solution

Hamiltonian

The Hamiltonian of the system is a function that describes the energy of the network. It is defined as the sum of the energies of all imbalanced triads.

Mean Field Approximation

The mean field approximation is a technique used to simplify the analysis of complex systems. It assumes that each node interacts with an average field created by all other nodes.

Bifurcation

A bifurcation is a point in the parameter space where the qualitative behavior of the system changes. In the context of structural balance, a bifurcation occurs at the critical temperature where the network transitions from a balanced to an imbalanced state.

Blue Sky Bifurcation and Saddle Node Bifurcation

A blue sky bifurcation is a type of bifurcation where a new stable solution appears. A saddle node bifurcation is a type of bifurcation where a stable and an unstable solution collide and disappear.

Eq1

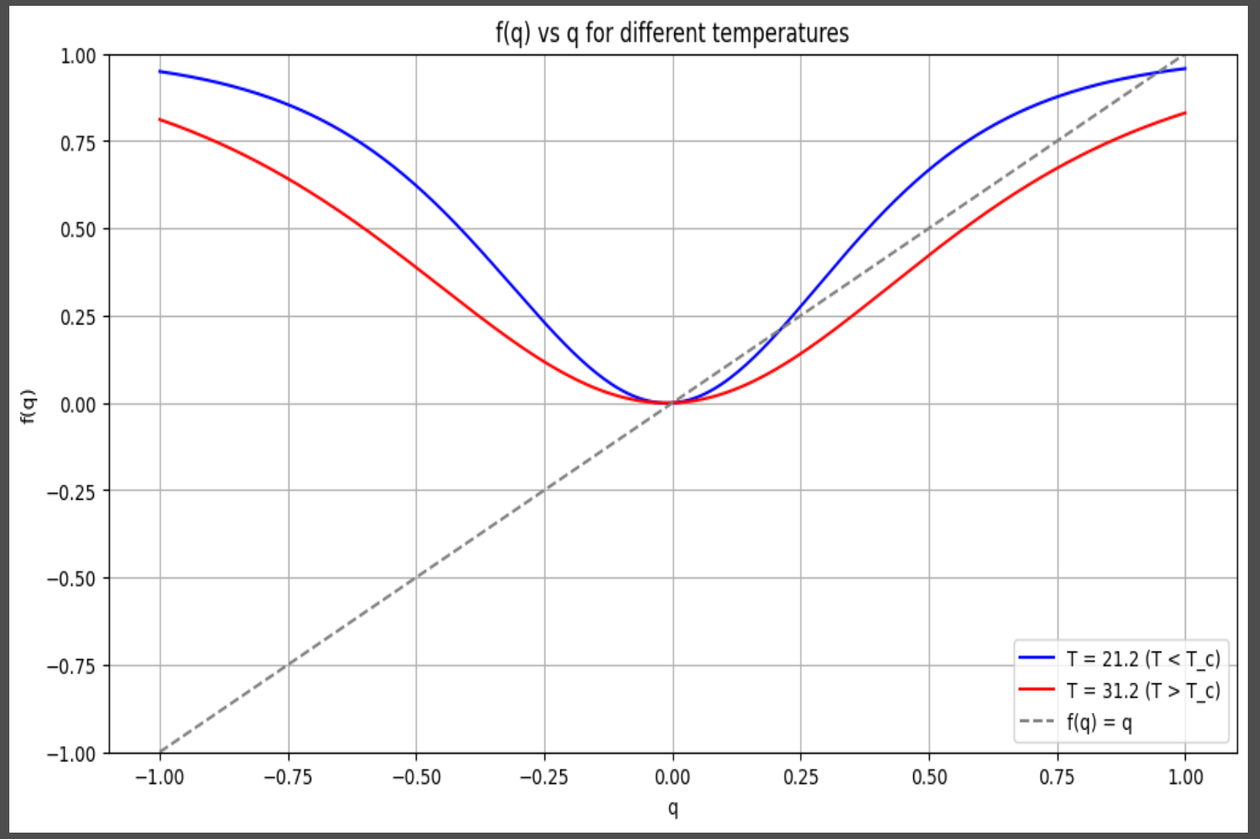


Fig. 3. Results simulated by me:
f(q) vs q for different tempratures

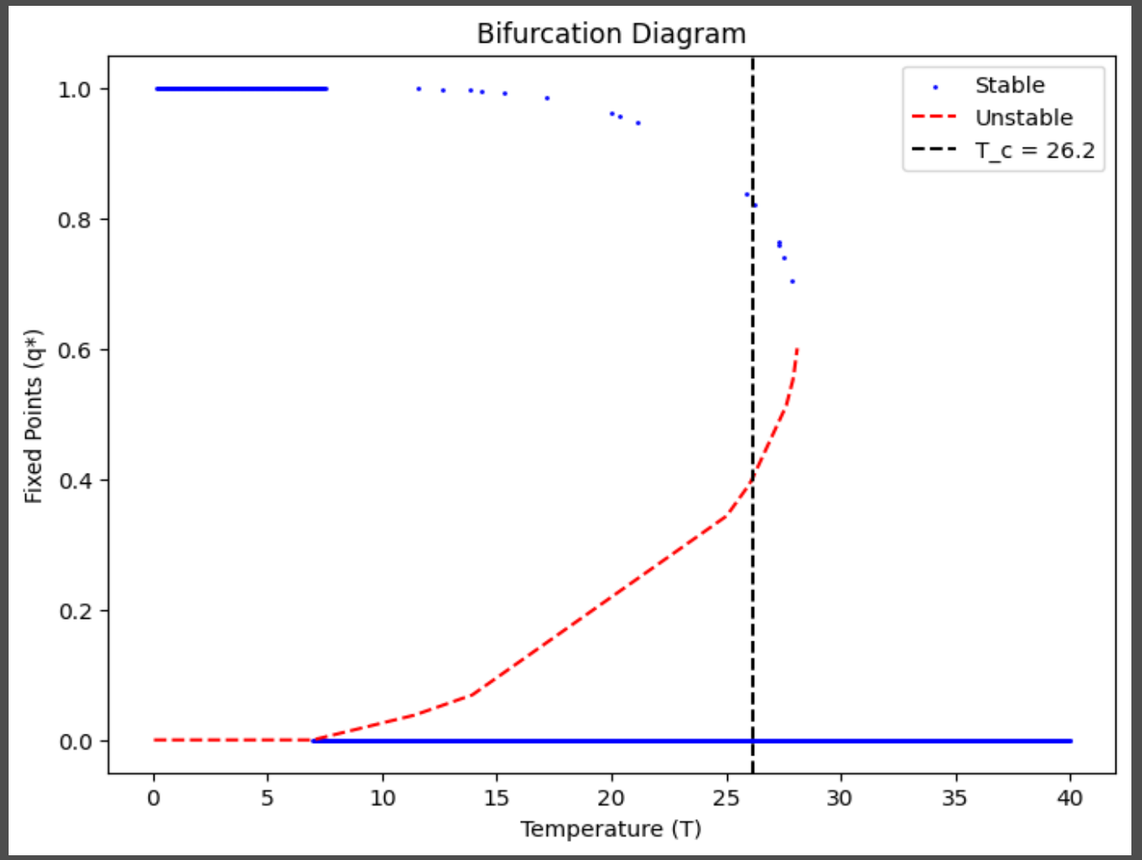


Fig. 4. Results simulated by me:
Bifurcation Diagram for showing stable and un-stable parts and considering critical temperature

Eq1

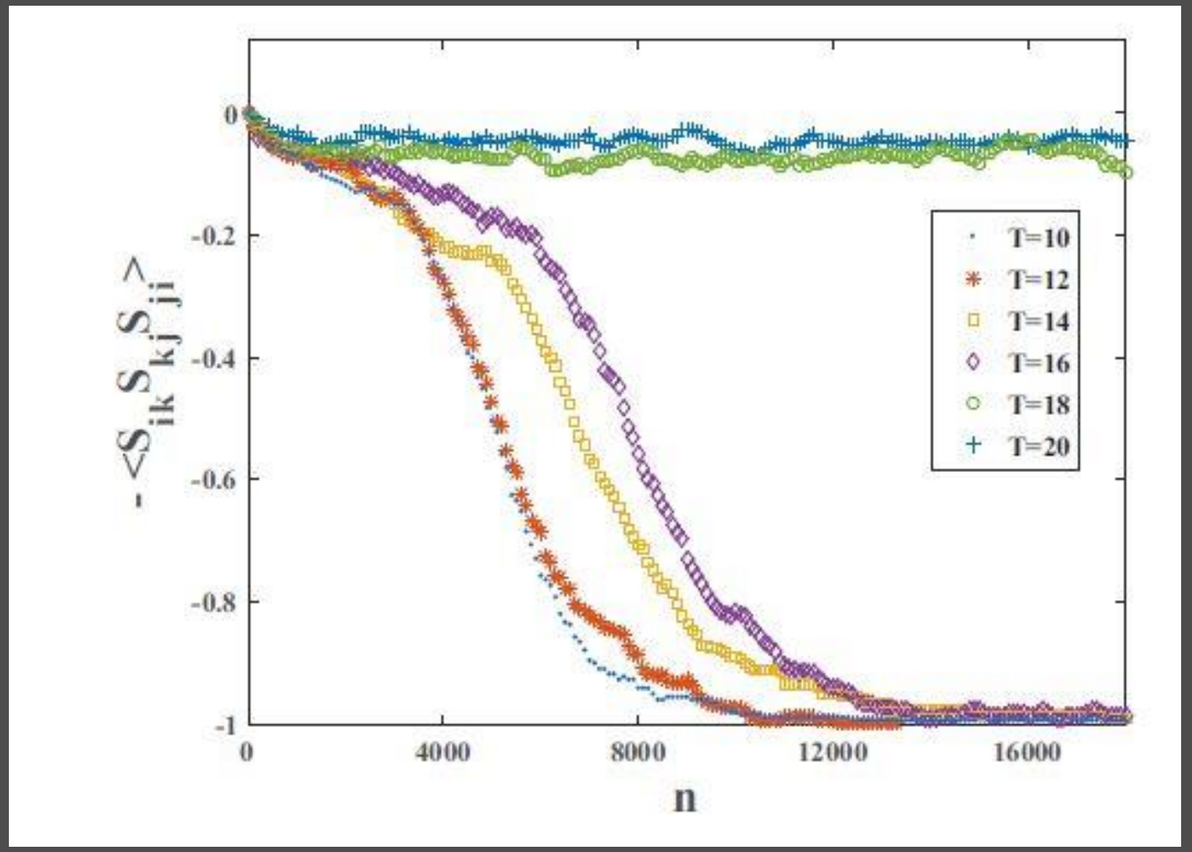


FIG. 5. The mean triad energy $-\langle S_{ik} S_{kj} S_{ji} \rangle$ vs Monte Carlo steps (n) for different values of temperature in a fully connected network with $N = 50$.

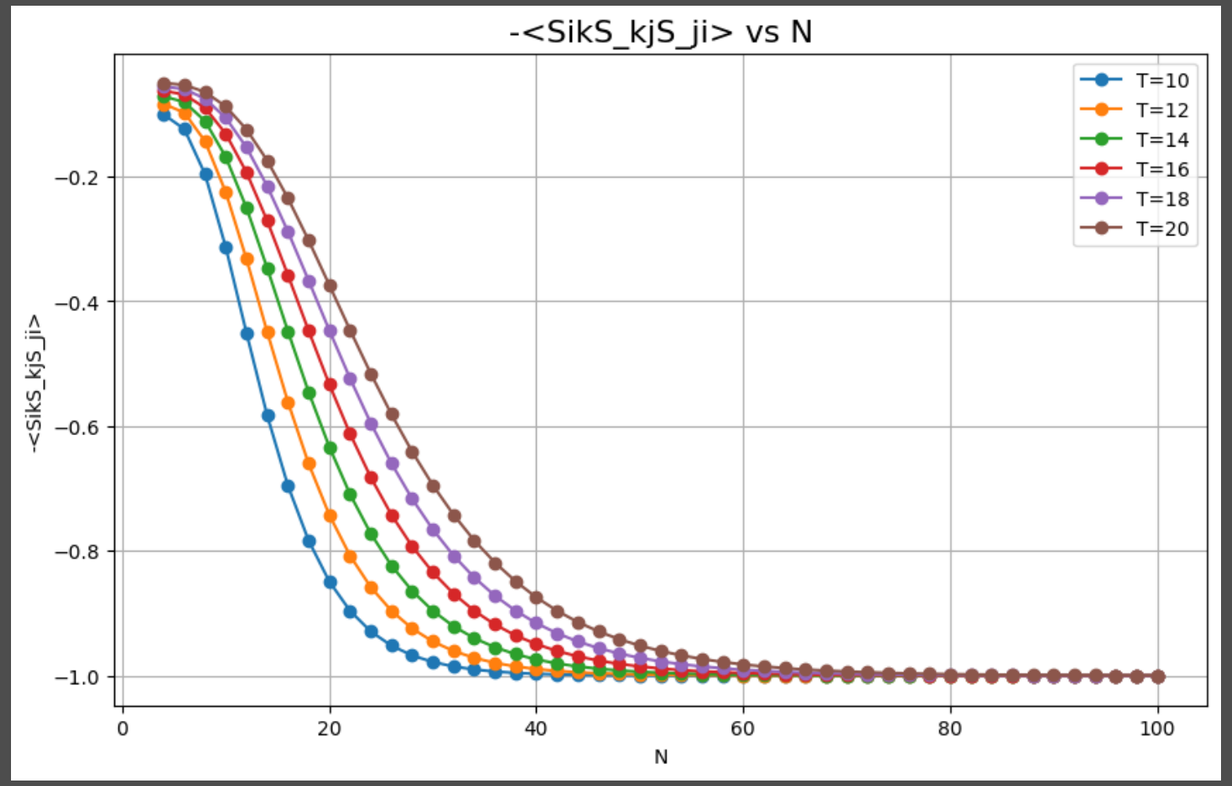


FIG. 6. Results simulated by me:
The mean triad energy vs N for different temperatures in range [10,20] with the step of 2

simulation and algorithm

1

Initialize Network

The simulation starts by initializing a network with a random configuration of positive and negative relationships.

2

Metropolis Algorithm

The Metropolis algorithm is a Monte Carlo method used to simulate the dynamics of the network. It involves randomly flipping links and accepting or rejecting the flip based on the energy change.

3

Energy Calculation

The energy of the network is calculated at each step of the simulation. It is defined as the sum of the energies of all imbalanced triads.

Eq1

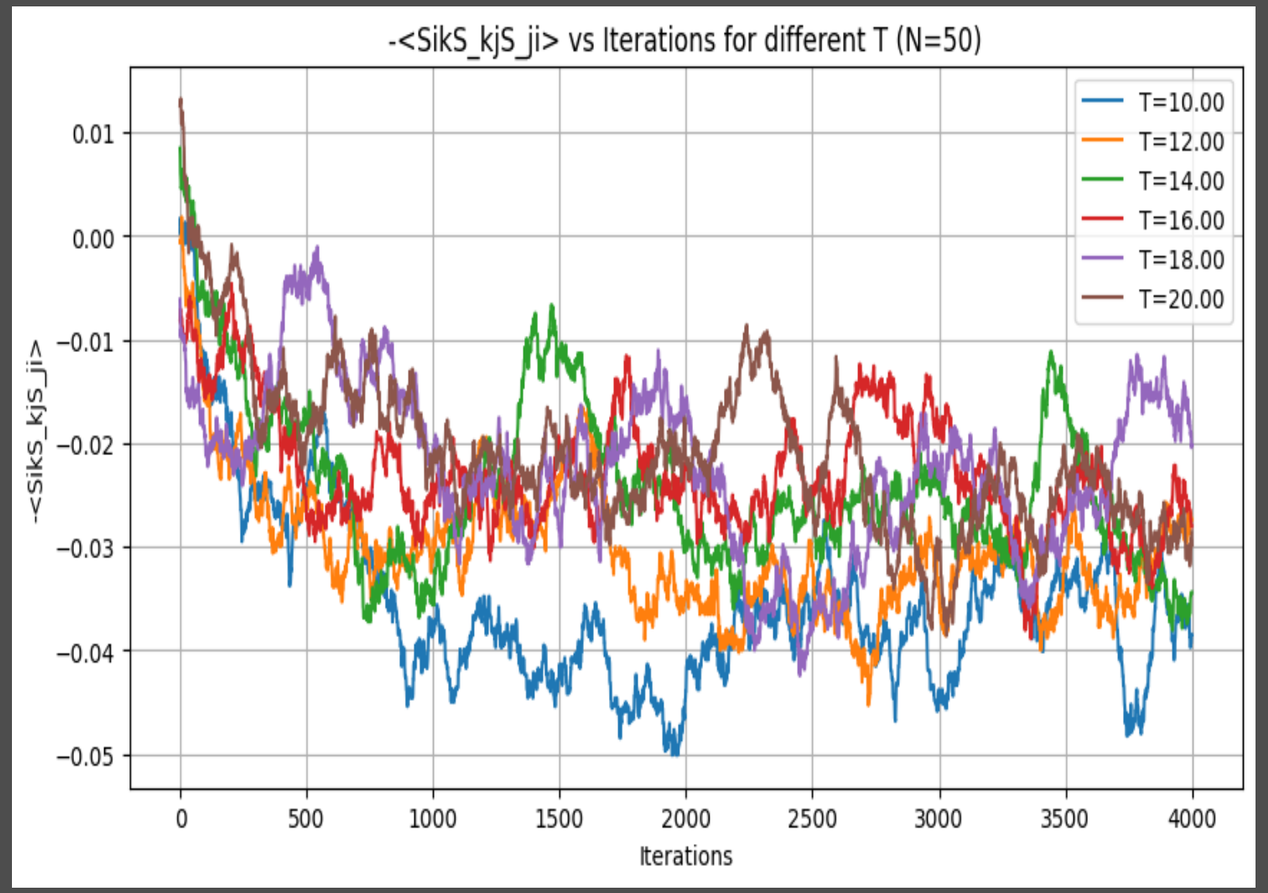


FIG. 7. Results simulated by me:
The mean triad energy vs number of iterations to 4000 monte carlos step for different temperatures in range [10,20] with the step of 2

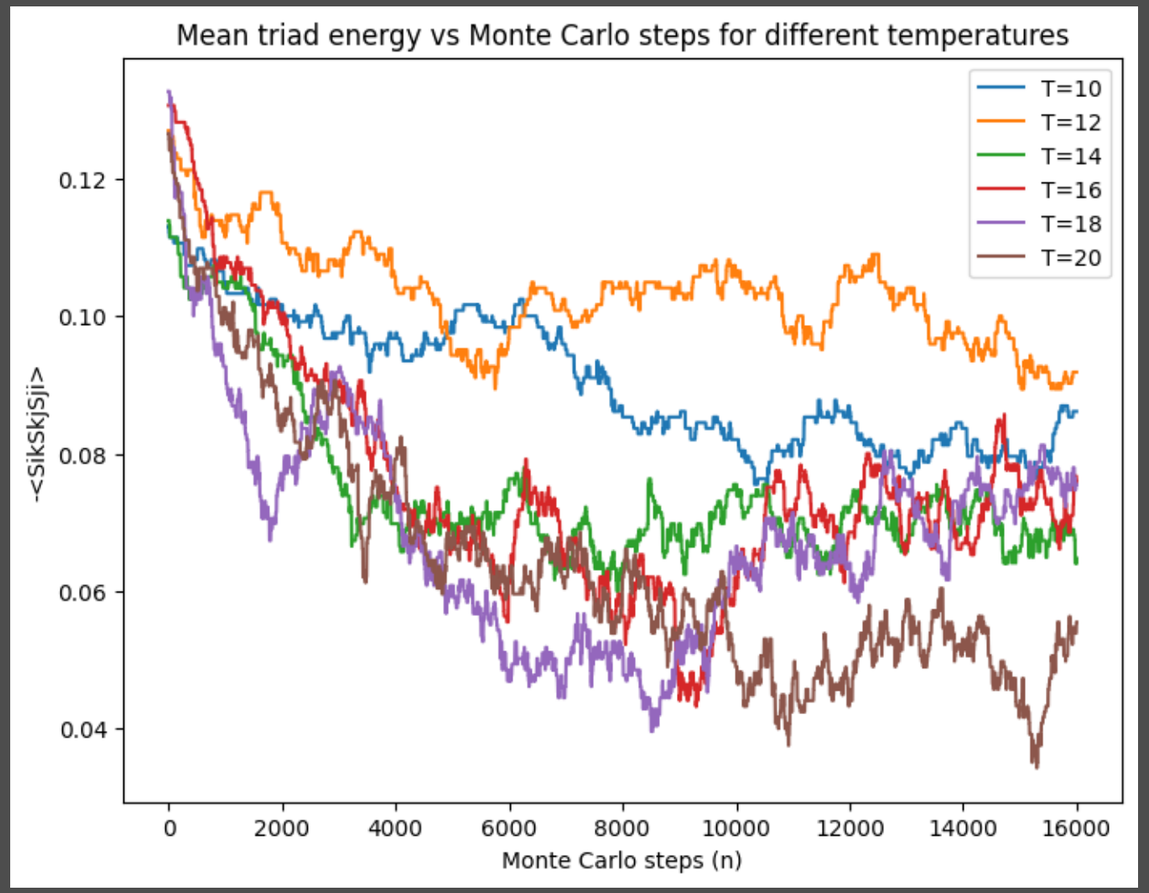


FIG. 8. Results simulated by me:
The mean triad energy vs vs number of iterations to 16000 monte carlos step for different temperatures in range [10,20] with the step of 2

simulation and algorithm

4

Link Flipping Criteria

The criteria for flipping a link are based on the energy change. If the flip reduces the energy, it is accepted. If it increases the energy, it is accepted with a probability that depends on the temperature.

5

Iterate for Various Temperatures

The simulation is run for various temperatures, allowing us to observe the impact of temperature on the network's evolution.

6

Analyze Results

The results of the simulation are analyzed to determine the critical temperature at which the network undergoes a phase transition from a balanced to an imbalanced state.

Eq1

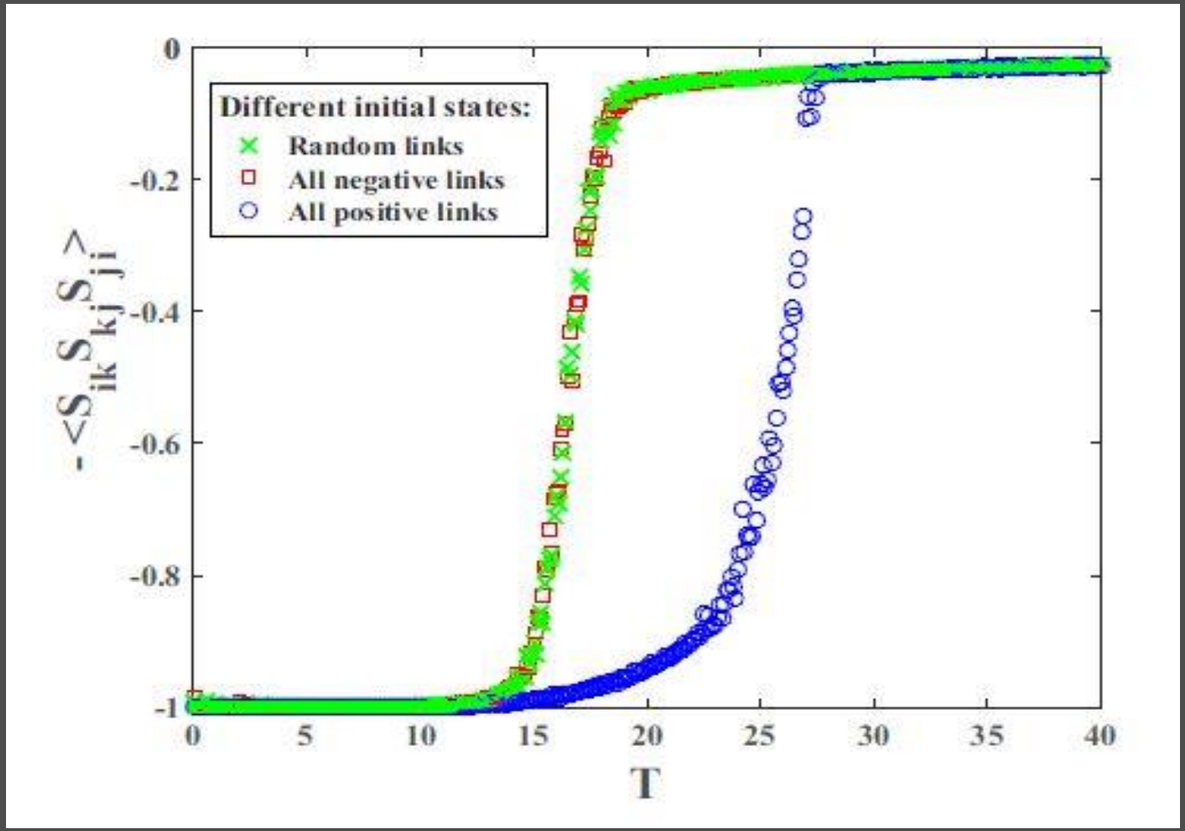


FIG. 9. The mean triad energy $-<Sik*Sk\,j*Sji>$ vs temperature (T) for different initial states differing in the percentage of balanced triads. We consider a balanced initial state with all positive links ($-Sik*Sk\,j*Sji = -1$) and an imbalanced initial state with all negative links ($-SikSk\,jSji = 1$). In a random initial state, we have almost equal positive and negative links ($-Sik*Sk\,j*Sji \sim 0$).

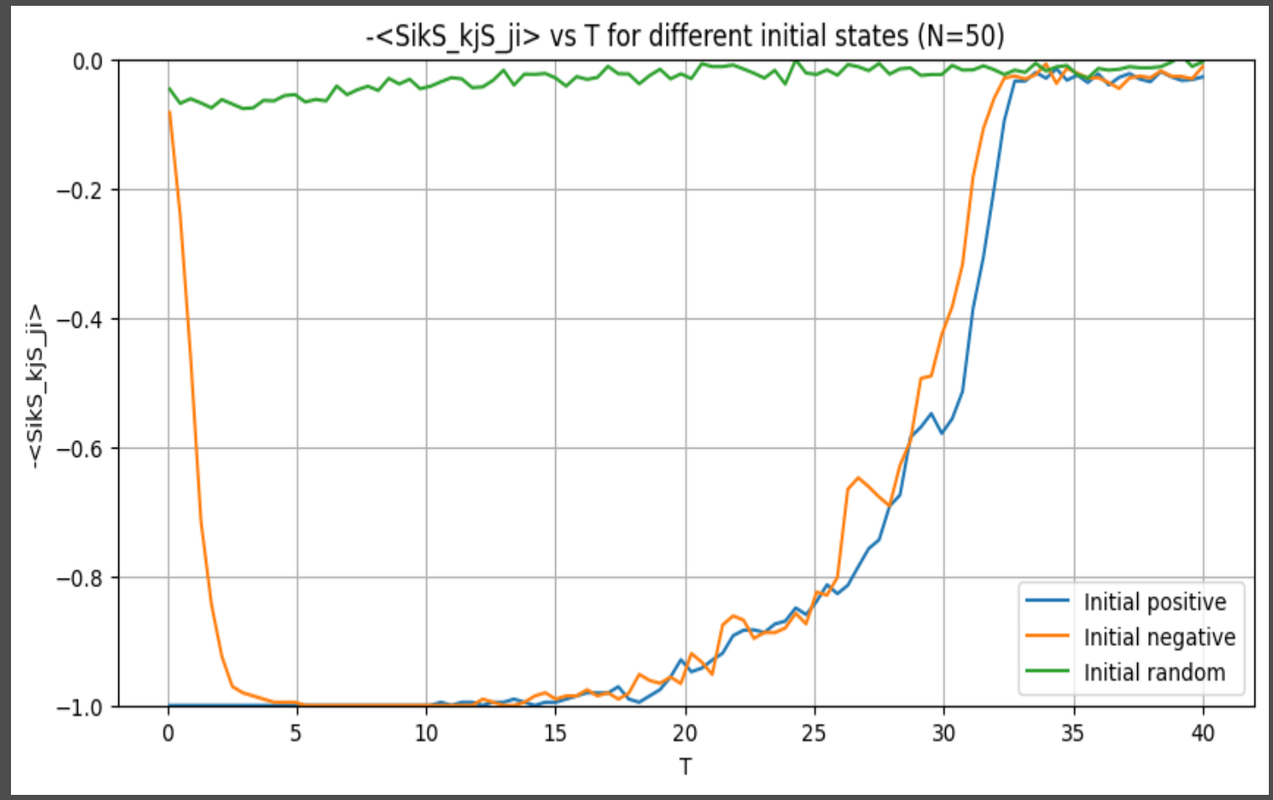


FIG. 10. Results simulated by me:
The mean triad energy vs T for 3 condition:

1. All + links
2. All – links
3. Random links

Eq1

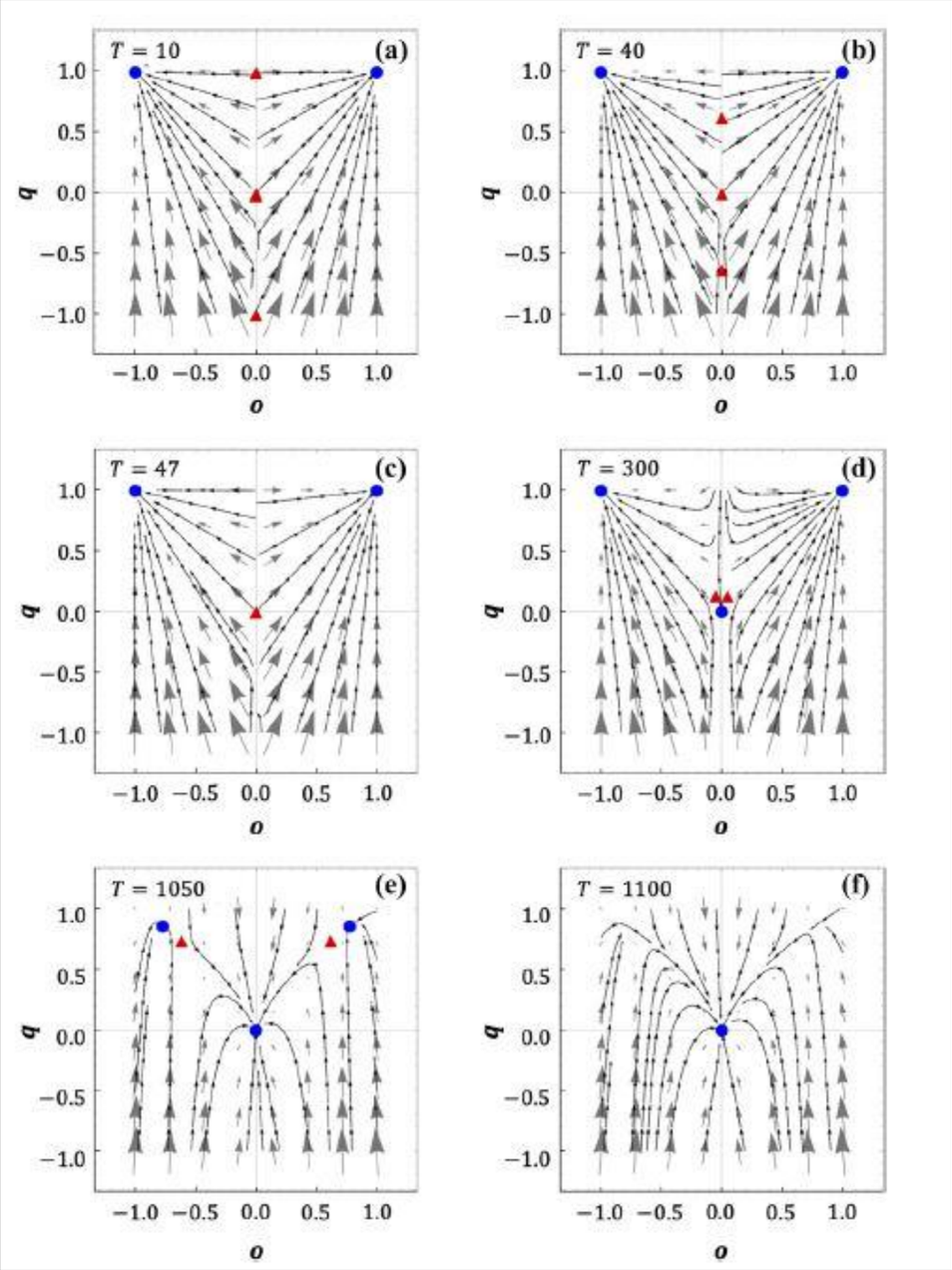
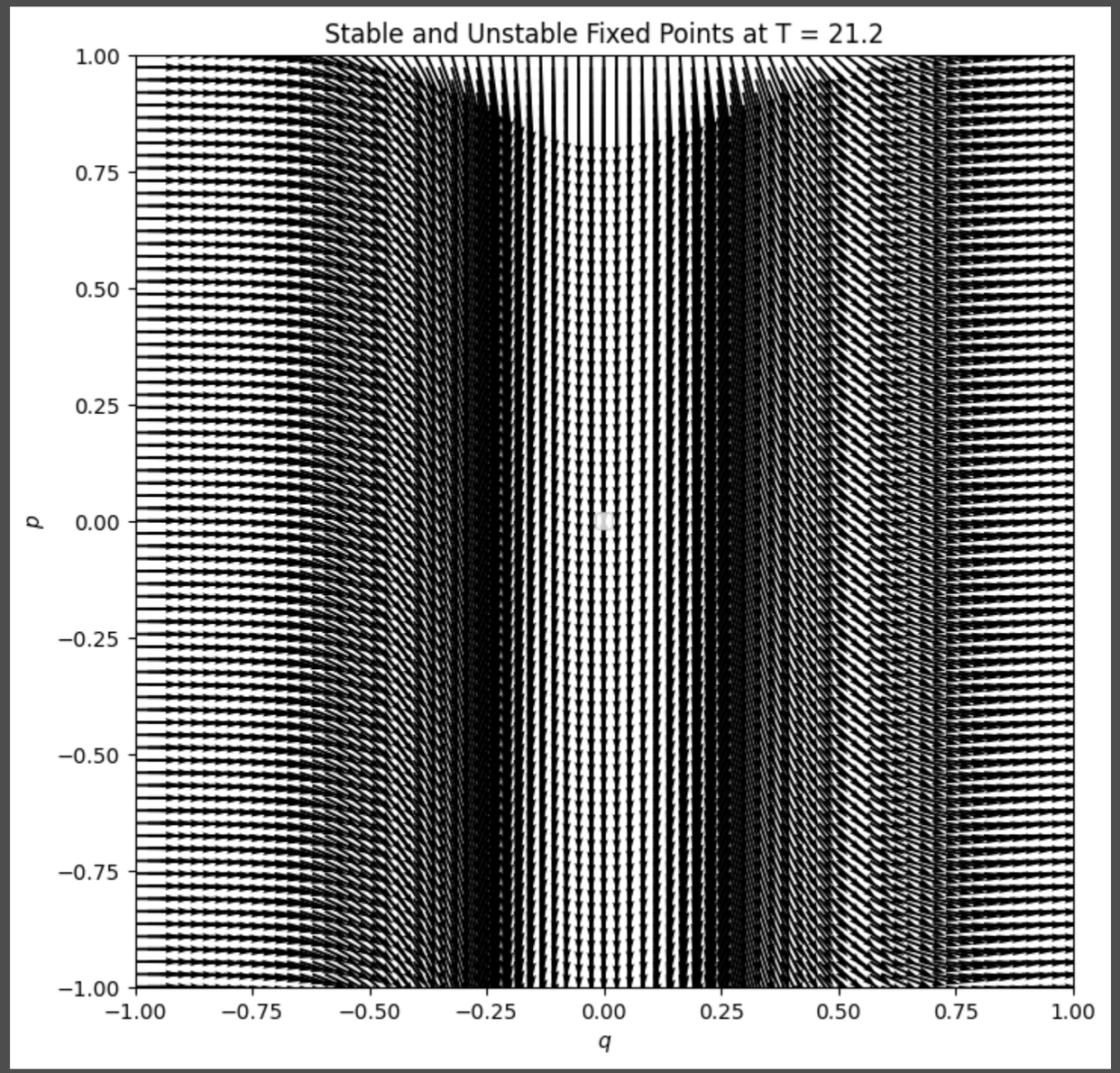


Fig. 11. Stable (blue circles) and unstable (red triangles) fixed points for different temperatures in the *q*-*o* plane. By increasing temperature, unstable fixed points move so close toward the trivial one in the origin [(a), (b), (c)], and they get away from it [(d), (e)]. When the temperature is bigger than *Tc* we have one stable fixed point (f). The number of nodes is 50.

Eq1



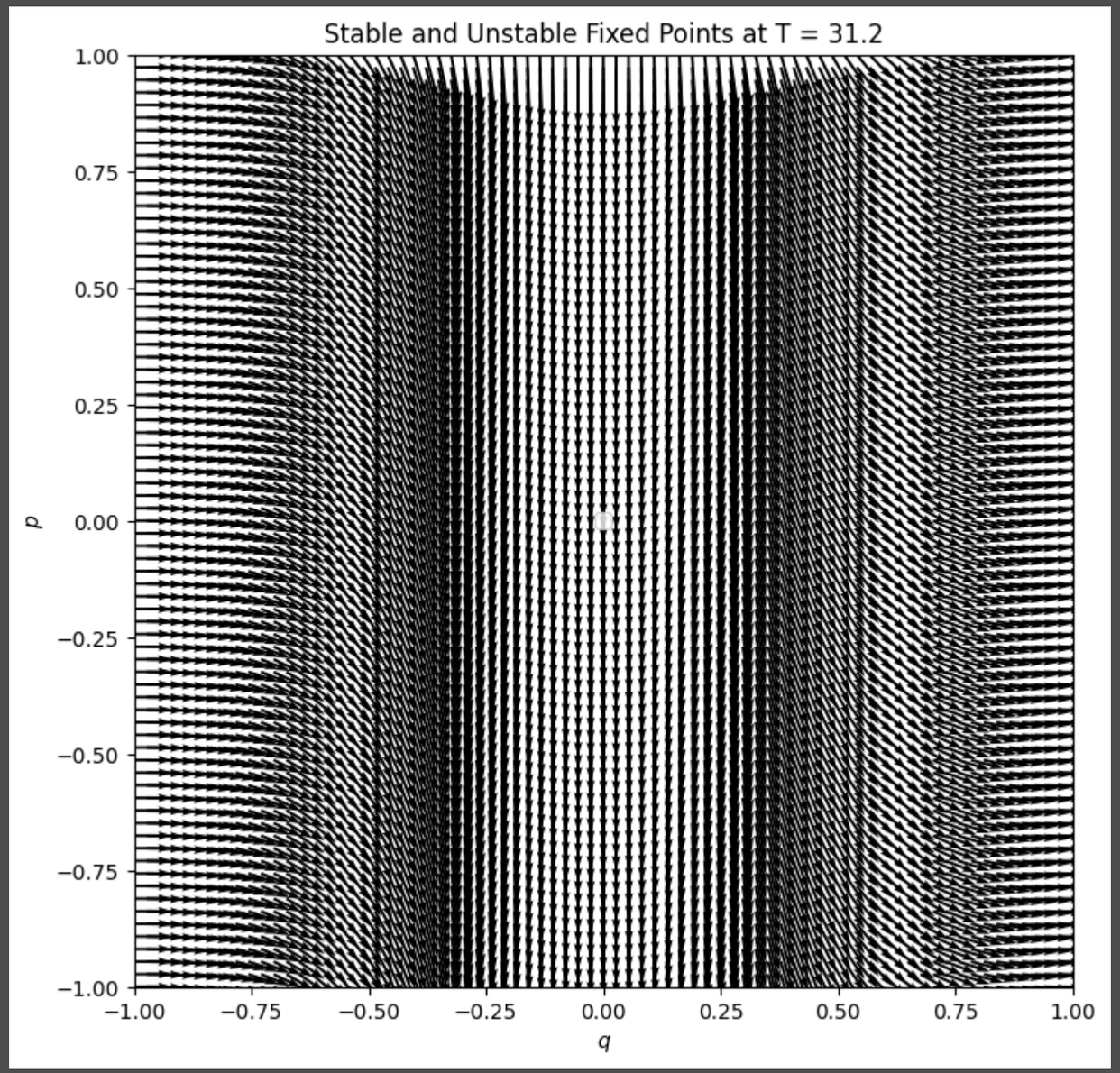


FIG. 12,13. Results simulated by me: Vector field of p vs q to show stable and unstable fixed points for T=21.2 (left) and T=31.2 (right) [based on Fig3 in second article]

Eq1

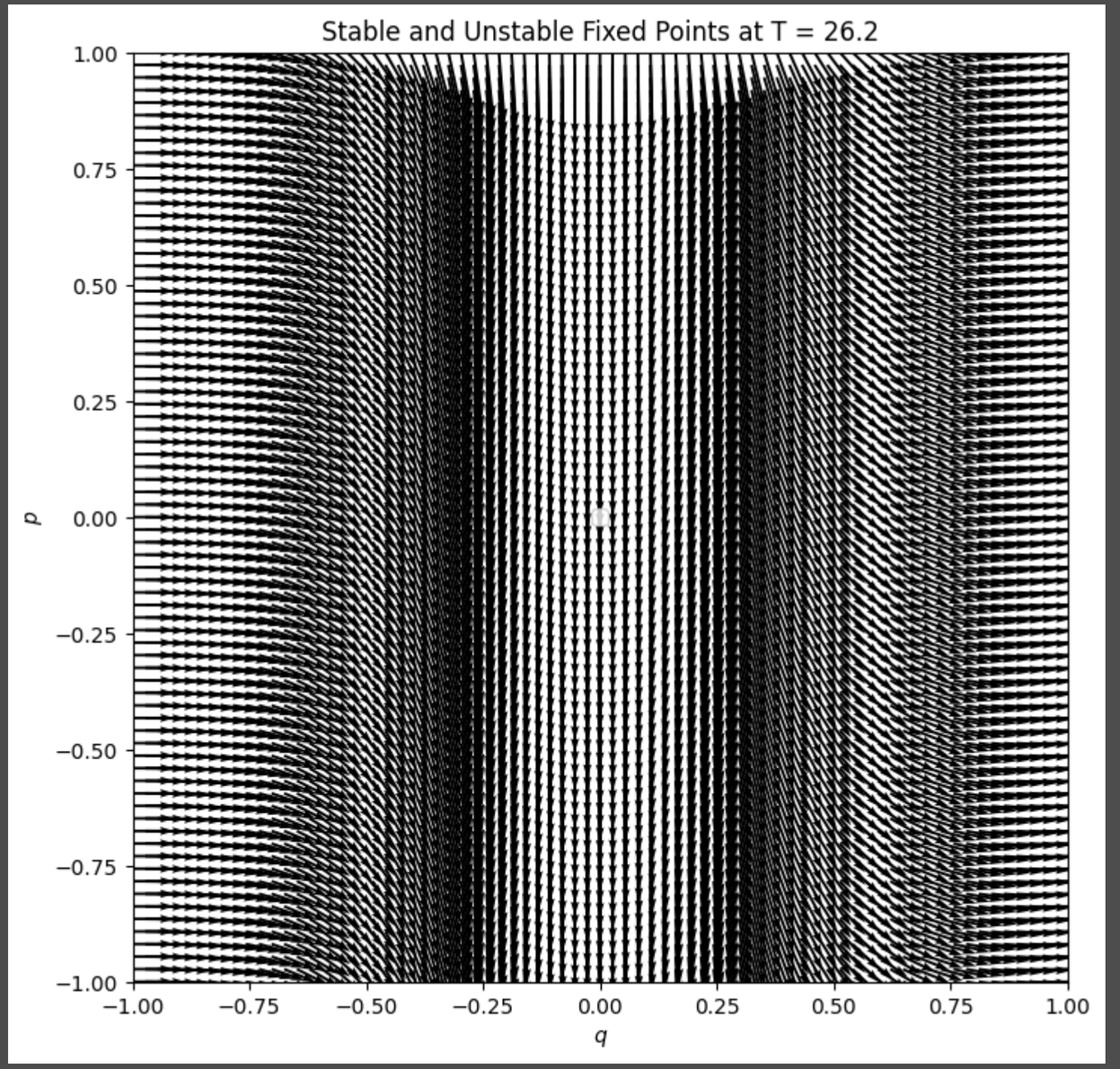


FIG. 14. Results simulated by me: Vector field of p vs q to show stable and unstable fixed points for critical Temperature

[based on Fig3 in second article]

conclusion



Impact of Temperature

Our analysis reveals that temperature plays a significant role in the stability and evolution of signed networks. As temperature increases, the network becomes more susceptible to tensions and conflicts, leading to an increase in the number of imbalanced triads.



Critical Threshold of Tolerance

There exists a critical threshold of tolerance, beyond which the network undergoes a phase transition and becomes dominated by imbalanced triads. This threshold is determined by the balance between the tendency towards balance and the influence of temperature.



Implications for Social Networks

These findings have important implications for understanding the dynamics of social networks. They suggest that the level of tolerance and flexibility in relationships can significantly impact the stability and evolution of these networks.

References

- Rabbani, F., Amir H. Shirazi, and G. R. Jafari. "Mean-field solution of structural balance dynamics in nonzero temperature." Physical Review E 99, no. 6 (2019): 062302.

- Kargaran, A., M. Ebrahimi, M. Riazi, A. Hosseiny, and G. R. Jafari. "Quartic balance theory: Global minimum with imbalanced triangles." Physical Review E 102, no. 1 (2020): 012310.

- https://www.youtube.com/watch?v=BBd68_q3Dgg (Bifurcations Part 1, Saddle-Node Bifurcation)
- https://www.youtube.com/watch?v=65ZKdscURXg (Bifurcations Part 2- Transcritical Bifurcation, Laser Model Example)